

# Wulff shapes

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Surface energies of the low-index  $(001)$ ,  $(011)$ , and  $(111)$  facets of Cu and Pd are computed from DFT symmetric slab models. For both metals the stability order is  $\gamma_{111} < \gamma_{001} < \gamma_{011}$ ; relaxation does not alter this sequence. The Wulff construction gives a truncated-octahedron for Cu ( $\{111\}$  64.6%,  $\{001\}$  23.7%,  $\{011\}$  11.7%) and a near-perfect octahedron for Pd ( $\{111\}$  89.4%,  $\{001\}$  10.6%,  $\{011\}$  absent). Interlayer relaxation is dominated by inward contraction, indicating attractive bulk bonding in both metals.

## I. INTRODUCTION

The equilibrium shape of a crystalline nanoparticle is determined by the anisotropy of its surface free energy. At fixed volume, the shape that minimises the total surface free energy is given by the Wulff construction, which relates the distance of each facet from the particle centre to the surface energy of that facet. [1] For fcc metals, the most relevant surfaces with low index are  $(001)$ ,  $(011)$ , and  $(111)$ . Their relative energies control whether the equilibrium morphology is close to an octahedron, a cube, or an intermediate polyhedron. [2, 3]

Copper and palladium are fcc transition metals of broad catalytic relevance. Cu-based nanoparticles are used in the water-gas shift and CO<sub>2</sub>-hydrogenation reactions, while Pd nanoparticles are key catalysts for selective hydrogenation and automotive exhaust-gas treatment. [5] Therefore, it is important to know which facets appear on the equilibrium shape because catalytic activity depends strongly on the exposed surface geometry.

In this exercise, we compute the fixed and relaxed surface energies of Cu and Pd from provided VASP calculations. Then, we construct the corresponding Wulff shapes in VESTA with these energies and quantify the fraction of exposed surface. Finally, as Bonus 1, the interlayer relaxation of the slab surfaces is analysed to determine whether the bulk bonding is attractive or repulsive.

## II. EMPLOYED MODELS & METHODOLOGY

### A. Slab calculations and surface energy expressions

The calculations were performed with the Vienna *ab initio* Simulation Package (VASP) using slab models built from previously optimised bulk Cu and Pd structures, with six atomic layers and the minimum vacuum required to separate the periodic slabs. [6] For each metal and facet, both the initial bulk-terminated slab and the fully relaxed slab are available. Because the slabs are symmetric, the two exposed surfaces are equivalent, and

the surface energy is obtained from

$$\gamma = \frac{E_{\text{slab}} - N\varepsilon_{\text{bulk}}}{2A}, \quad (1)$$

where  $E_{\text{slab}}$  is the total slab energy,  $N$  is the number of atoms in the slab,  $\varepsilon_{\text{bulk}}$  is the bulk energy per atom, and  $A$  is the area of one surface. [4]

The *fixed* surface energy is calculated from the unrelaxed slab energy,

$$\gamma_{\text{fix}} = \frac{E_{\text{slab}}^{\text{fix}} - N\varepsilon_{\text{bulk}}}{2A}, \quad (2)$$

while the *relaxed* surface energy is obtained from the final optimised slab energy,

$$\gamma_{\text{relax}} = \frac{E_{\text{slab}}^{\text{relax}} - N\varepsilon_{\text{bulk}}}{2A}. \quad (3)$$

The relaxation contribution is defined as

$$\Delta\gamma = \gamma_{\text{relax}} - \gamma_{\text{fix}}. \quad (4)$$

In most cases  $\Delta\gamma < 0$ , which indicates a stabilisation by ionic relaxation, although very small positive values may also appear within the numerical accuracy of the calculation.

### B. Wulff construction

The Wulff theorem states that the equilibrium shape is obtained by placing planes with Miller index  $\{hkl\}$  at distances  $h_{hkl}$  from the centre proportional to their surface energies,

$$\frac{\gamma_{001}}{h_{001}} = \frac{\gamma_{011}}{h_{011}} = \frac{\gamma_{111}}{h_{111}} = \text{constant}. \quad (5)$$

In practice, the construction was performed in VESTA by normalising all distances to the most stable facet,

$$h_{hkl} = \frac{\gamma_{hkl}}{\gamma_{\min}}, \quad (6)$$

with  $\gamma_{\min} = \gamma_{111}$  for both Cu and Pd. The corresponding planes were introduced in the *Crystal Shape* module of VESTA, and the exposed areas of each facet family were read from the program output. [7]

### C. Bonus 1: interlayer relaxation

For Bonus 1, the interlayer spacings were extracted from the atomic coordinates in the slab POSCAR/CONTCAR files. For each surface, the first and second interlayer distances ( $d_{12}$  and  $d_{23}$ ) were computed before and after relaxation, and compared with the ideal bulk interlayer spacing  $d_{\text{bulk}}$  expected for an fcc crystal:

$$d_{\text{bulk}}^{(001)} = \frac{a}{2}, \quad d_{\text{bulk}}^{(011)} = \frac{a}{2\sqrt{2}}, \quad d_{\text{bulk}}^{(111)} = \frac{a}{\sqrt{3}}, \quad (7)$$

where  $a$  is the cubic lattice parameter. The relative interlayer relaxation is defined as

$$\Delta d_{ij}(\%) = \frac{d_{ij}^{\text{relax}} - d_{ij}^{\text{fix}}}{d_{\text{bulk}}} \times 100. \quad (8)$$

Negative values correspond to inward contraction of the slab layers, and positive values, to expansion.

### D. Computational details

The bulk reference energies were extracted from the OUTCAR of a conventional four-atom fcc cell:

$$\varepsilon_{\text{bulk}}^{\text{Cu}} = -3.725780 \text{ eV atom}^{-1}$$

and

$$\varepsilon_{\text{bulk}}^{\text{Pd}} = -5.176240 \text{ eV atom}^{-1}.$$

The lattice parameters are

$$a_{\text{Cu}} = 3.61473 \text{ \AA}$$

and

$$a_{\text{Pd}} = 3.89333 \text{ \AA}.$$

The analysis is done with Python scripts available at the GitHub repository linked in this report.

## III. RESULTS AND DISCUSSION

### A. Step 1: Surface energies of Cu and Pd

#### 1. Fixed and relaxed surface energies

Table I lists the fixed and relaxed surface energies of the six slab systems studied. For both Cu and Pd the stability ordering is

$$\gamma_{111} < \gamma_{001} < \gamma_{011}, \quad (9)$$

which is the expected sequence for fcc metals and reflects the fact that  $(111)$  is the close-packed surface with the smallest number of broken bonds per surface atom. [2, 3]

TABLE I: Fixed ( $\gamma_{\text{fix}}$ ) and relaxed ( $\gamma_{\text{relax}}$ ) surface energies, and the relaxation contribution  $\Delta\gamma = \gamma_{\text{relax}} - \gamma_{\text{fix}}$ , for the Cu and Pd facets. All values are in  $\text{J m}^{-2}$ . The most stable facet for each metal is  $(111)$ .

| Metal | Surface | $\gamma_{\text{fix}}$ | $\gamma_{\text{relax}}$ | $\Delta\gamma$ |
|-------|---------|-----------------------|-------------------------|----------------|
| Cu    | (001)   | 1.5190                | 1.5102                  | -0.0088        |
| Cu    | (011)   | 1.5851                | 1.5421                  | -0.0430        |
| Cu    | (111)   | 1.3314                | 1.3313                  | -0.0001        |
| Pd    | (001)   | 1.5044                | 1.5022                  | -0.0022        |
| Pd    | (011)   | 1.5871                | 1.5406                  | -0.0465        |
| Pd    | (111)   | 1.1333                | 1.1352                  | +0.0019        |

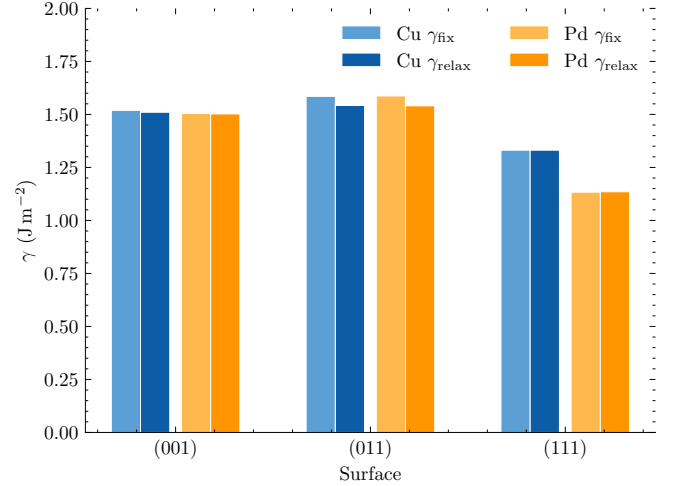


FIG. 1: Fixed ( $\gamma_{\text{fix}}$ ) and relaxed ( $\gamma_{\text{relax}}$ ) surface energies of Cu (blue) and Pd (orange) for the three low-index facets.

The most stable surface is  $(111)$  for both metals. The  $(111)$  plane is the close-packed surface of the fcc lattice, where each surface atom has the maximum number of nearest neighbours, so the density of broken bonds is lowest and the surface energy is smallest. Then,  $(011)$  is the most open surface and carries the largest energy penalty.

We should highlight that the surface stability is determined by the absolute value of  $\gamma_{\text{relax}}$ , not by the sign of  $\Delta\gamma$ . The small positive value  $\Delta\gamma = +0.0019 \text{ J m}^{-2}$  for Pd $(111)$  does not mean this surface is unstable or that relaxation destabilises it. Pd $(111)$  has the lowest  $\gamma_{\text{relax}} = 1.135 \text{ J m}^{-2}$  of all systems, making it the most stable surface overall. The positive  $\Delta\gamma$  simply reflects the fact that the  $(111)$  geometry is already very close to the relaxed optimum, so the ionic displacements are negligible and the associated change in energy lies within the numerical noise of the calculation.

The difference between Cu and Pd is especially marked for  $(111)$ :  $\gamma_{\text{relax}}^{\text{Pd}} = 1.135 \text{ J m}^{-2}$  is  $\approx 0.20 \text{ J m}^{-2}$  lower than  $\gamma_{\text{relax}}^{\text{Cu}} = 1.331 \text{ J m}^{-2}$ . There is a bigger  $\{111\}$  dominance in the Pd Wulff shape.

## 2. Effect of ionic relaxation

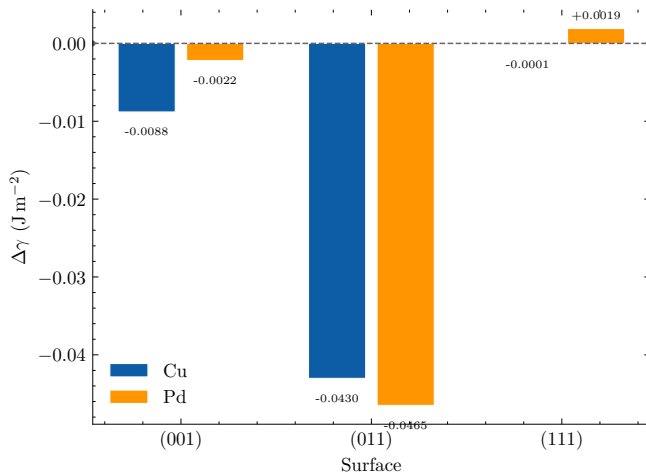


FIG. 2: Relaxation contribution  $\Delta\gamma = \gamma_{\text{relax}} - \gamma_{\text{fix}}$  for Cu (blue) and Pd (orange). Negative values indicate stabilisation by ionic relaxation. The open  $(011)$  facet relaxes the most, whereas the  $(111)$  facet is only weakly affected.

The effect of ionic relaxation depends on the surface geometry. For both metals the largest stabilisation occurs at  $(011)$ :  $\Delta\gamma = -0.043 \text{ J m}^{-2}$  for Cu and  $-0.047 \text{ J m}^{-2}$  for Pd. The  $(001)$  surface relaxes less and the  $(111)$  surface is almost unchanged. The positive value for Pd $(111)$  ( $+0.0019 \text{ J m}^{-2}$ ) is best interpreted as numerical noise, as discussed above.

Relaxation is therefore not a determining factor in the surface stability. The ordering  $\gamma_{111} < \gamma_{001} < \gamma_{011}$  stays the same for fixed and relaxed energies in both metals. Even though the effect on  $(011)$  is significant ( $\sim 3\%$  of the surface energy), it is not large enough to modify the stability sequence, so the same facet ranking is obtained with or without ionic relaxation.

3. Wulff  $h$ -values

Table II contains the normalised Wulff  $h$ -values obtained from the relaxed surface energies. These are the values entered in VESTA to construct the equilibrium shape.

TABLE II: Normalised Wulff  $h$ -values computed from the relaxed surface energies,  $h_{hkl} = \gamma_{hkl}/\gamma_{111}$ , with  $h_{111} = 1$ .

| Metal | $h_{001}$ | $h_{011}$ | $h_{111}$ |
|-------|-----------|-----------|-----------|
| Cu    | 1.1344    | 1.1584    | 1.0000    |
| Pd    | 1.3187    | 1.3527    | 1.0000    |

For Cu,  $h_{001}$  and  $h_{011}$  are only slightly above unity, which means that all three facet families remain geometrically competitive and all three are expected to appear

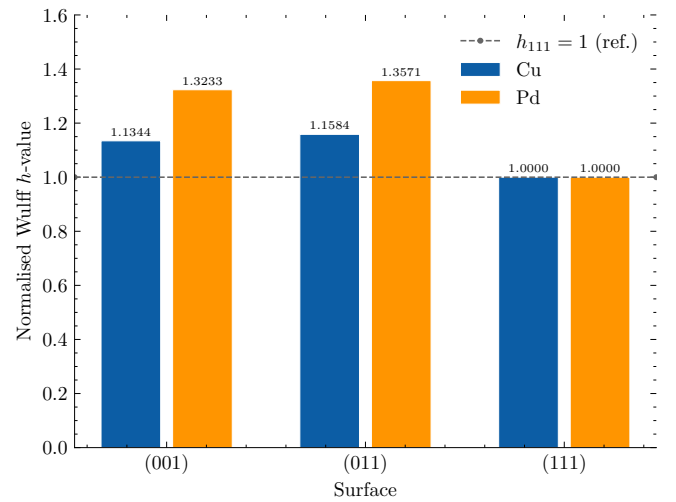


FIG. 3: Normalised Wulff  $h$ -values ( $h_{111} = 1$ ) for Cu (blue) and Pd (orange).

in the final shape. For Pd, both values are much larger, so the  $\{001\}$  and  $\{011\}$  planes will be further from the centre of the Wulff polyhedron, having a less exposed area and a shape dominated almost entirely by  $\{111\}$  faces.

## B. Step 2: Wulff shapes of Cu and Pd nanoparticles

## 1. Cu Wulff shape

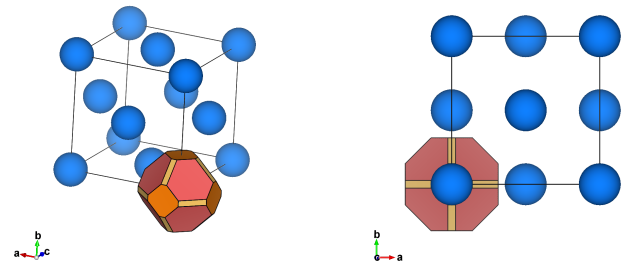


FIG. 4: Equilibrium Wulff shape of Cu generated in VESTA from the relaxed surface energies: isometric view (left) and view along  $[001]$  (right). The morphology is a truncated-octahedron-like fcc polyhedron dominated by large  $\{111\}$  facets, with smaller square  $\{001\}$  truncations and narrow residual  $\{011\}$  facets.

The Cu nanoparticle Wulff shape is the expected truncated octahedron-like shape, like most fcc metals. The  $\{111\}$  faces are the largest, with  $\{001\}$  truncations. There is also a small  $\{011\}$  contribution, because the  $(011)$  surface is only slightly less stable than the  $(001)$ .

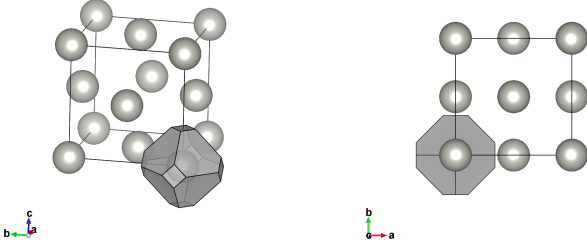


FIG. 5: Equilibrium Wulff shape of Pd generated in VESTA from the relaxed surface energies: isometric view (left) and view along  $[001]$  (right). The morphology is a near-perfect octahedron dominated by  $\{111\}$  facets, with only small  $\{001\}$  truncations and no  $\{011\}$  facet area.

## 2. Pd Wulff shape

The Pd shape is different from Cu, much closer to a regular octahedron and is very dominated by  $\{111\}$  faces. The  $\{011\}$  contributions are not there, as the large Wulff anisotropy predicts. Since the Pd  $(111)$  surface energy is far lower than the other two, the  $\{111\}$  family occupies nearly the entire equilibrium surface area.

TABLE III: Exposed surface fractions of the Wulff equilibrium shapes obtained from the VESTA face-area output.

| Metal | $\{111\}$ (%) | $\{001\}$ (%) | $\{011\}$ (%) |
|-------|---------------|---------------|---------------|
| Cu    | 64.57         | 23.74         | 11.70         |
| Pd    | 89.37         | 10.63         | 0.00          |

In table III, we see the percentage of exposed surfaces. Cu has all three families and has more types of facets, while Pd only has  $\{111\}$  and  $\{001\}$ .

## C. Bonus 1: Interlayer relaxation distances

Table IV shows the first interlayer relaxation for all six slab systems, with the ideal bulk spacing  $d_{\text{bulk}}$  obtained from the cubic lattice constant. The behaviour is consistent with the previous surface energy analysis: the most open surface,  $(011)$ , shows by far the largest inward contraction, while the close-packed  $(111)$  surface changes least.

For Cu, all three surfaces contract inward. The strongest contraction is for  $(011)$  ( $-10.26\%$ ), and then for  $(001)$  ( $-2.37\%$ ), whereas  $(111)$  is almost unchanged ( $-0.83\%$ ). Therefore, the more open the surface, the more the under coordinated atoms contract inward to recover lost bonding.

Pd shows the same behaviour for  $(001)$  and  $(011)$ , with contractions of  $-1.18\%$  and  $-9.69\%$ . The only exception is Pd $(111)$ , which shows a tiny expansion of  $+0.07\%$ . Nevertheless, this is small enough to be

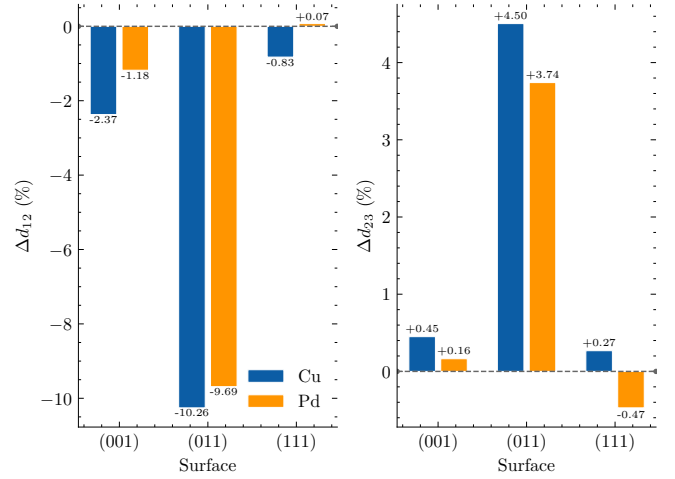


FIG. 6: Bonus 1: relative first-layer relaxation  $\Delta d_{12}$  for Cu (blue) and Pd (orange). The largest inward contraction is found for the open  $(011)$  surface, whereas  $(111)$  is almost unchanged. The tiny Pd $(111)$  expansion ( $+0.07\%$ ) is negligible.

considered a numerical uncertainty, as happened with  $\Delta\gamma = +0.0019 \text{ J m}^{-2}$ .

Therefore, we can say that Cu and Pd bulk bonding is *attractive*. The dominant relaxation response is inward contraction: once the outer neighbours are removed, the surface atoms are pulled toward the bulk to strengthen their remaining bonds. This is in agreement with the previous surface energy analysis.

## IV. CONCLUSIONS

For both Cu and Pd the DFT slab calculations give the stability sequence  $\gamma_{111} < \gamma_{001} < \gamma_{011}$ . Ionic relaxation lowers mainly the  $(011)$  surface energy and is negligible for  $(111)$ . It does not alter the ordering and is not a determining factor in surface stability.

The Wulff constructions give two distinct morphologies. Cu is a truncated octahedron having  $\{111\}$  (64.6%),  $\{001\}$  (23.7%), and  $\{011\}$  (11.7%). Pd is a near octahedron with  $\{111\}$  (89.4%) and small  $\{001\}$  truncations (10.6%) only, with  $\{011\}$  absent. The difference is because the much larger surface energy anisotropy of Pd: the  $\{111\}$  preference is strong enough to suppress  $\{011\}$  entirely from the equilibrium shape. As a consequence, catalytic studies of Cu must consider all three low-index facets, while for Pd only  $\{111\}$  and  $\{001\}$  are relevant under equilibrium conditions.

Additionally, Bonus 1 shows that the dominant interlayer response is inward contraction, largest for  $(011)$  and negligible for  $(111)$ , confirming attractive bulk bonding in both metals.

TABLE IV: Bonus 1: ideal bulk interlayer spacing  $d_{\text{bulk}}$ , first-layer spacing before and after relaxation, and relative interlayer relaxation  $\Delta d_{12} = (d_{12}^{\text{relax}} - d_{12}^{\text{fix}})/d_{\text{bulk}}$ . All distances in Å.

| Metal | Surface | $d_{\text{bulk}}$ | $d_{12}^{\text{fix}}$ | $d_{12}^{\text{relax}}$ | $\Delta d_{12}$ (%) |
|-------|---------|-------------------|-----------------------|-------------------------|---------------------|
| Cu    | (001)   | 1.8074            | 1.8157                | 1.7728                  | −2.37               |
| Cu    | (011)   | 1.2780            | 1.2839                | 1.1528                  | −10.26              |
| Cu    | (111)   | 2.0870            | 2.0950                | 2.0778                  | −0.83               |
| Pd    | (001)   | 1.9467            | 1.9755                | 1.9525                  | −1.18               |
| Pd    | (011)   | 1.3765            | 1.3969                | 1.2635                  | −9.69               |
| Pd    | (111)   | 2.2478            | 2.2823                | 2.2839                  | +0.07               |

**Code:** GitHub repository

<https://github.com/itxasoma/V2--Wulff-Shapes>

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